

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 03



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Recap of Previous Lecture



Topic

Number of subsets of a given set



Topic

Power set

$$|P(A)| = 2^{|A|}$$



Topic

Venn diagram



Topic

Set operations



Topics to be Covered



Topic

Set operations and properties of set operations



Topic

Multi-set



Topic

Cartesian product



Topic

Relation



$$(A-B) \cap (B-A) = \emptyset \quad \text{always}$$

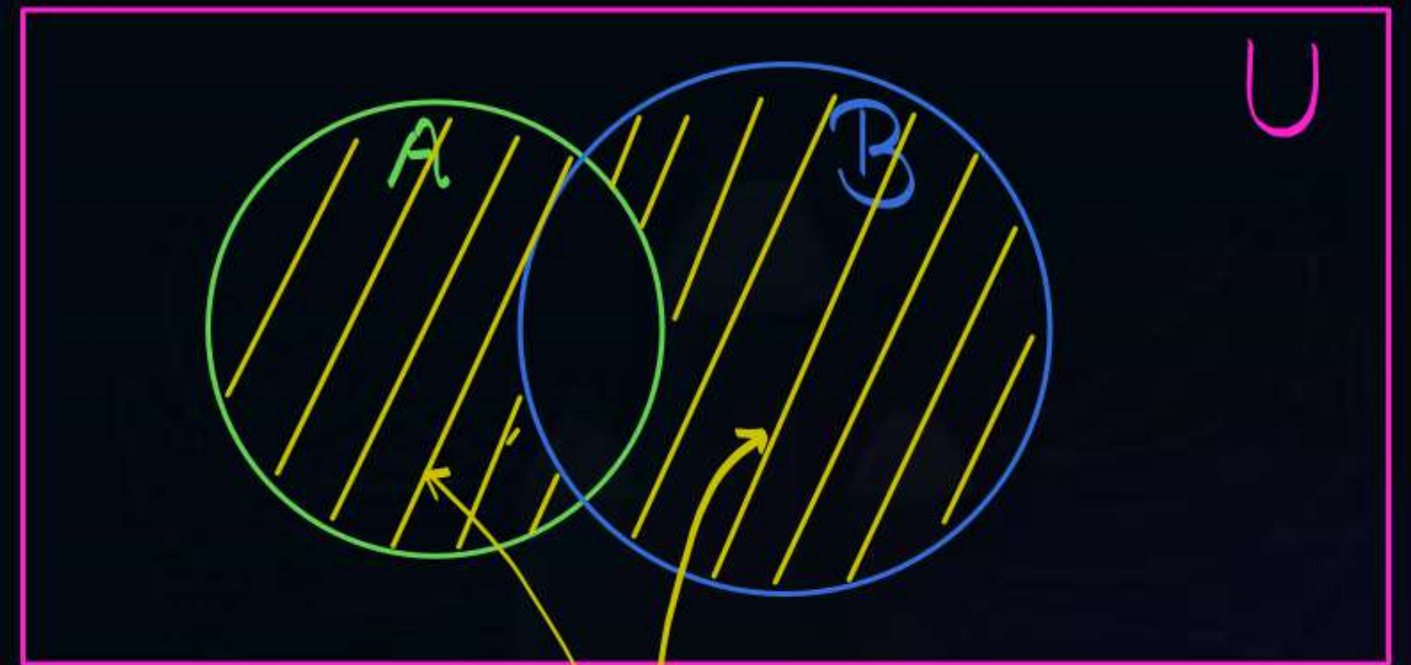


Topic : Symmetric difference of two sets

Let A and B are two sets,

$$A \Delta B = \{x \mid (x \in A \text{ or } x \in B) \text{ and } x \notin A \cap B\}$$

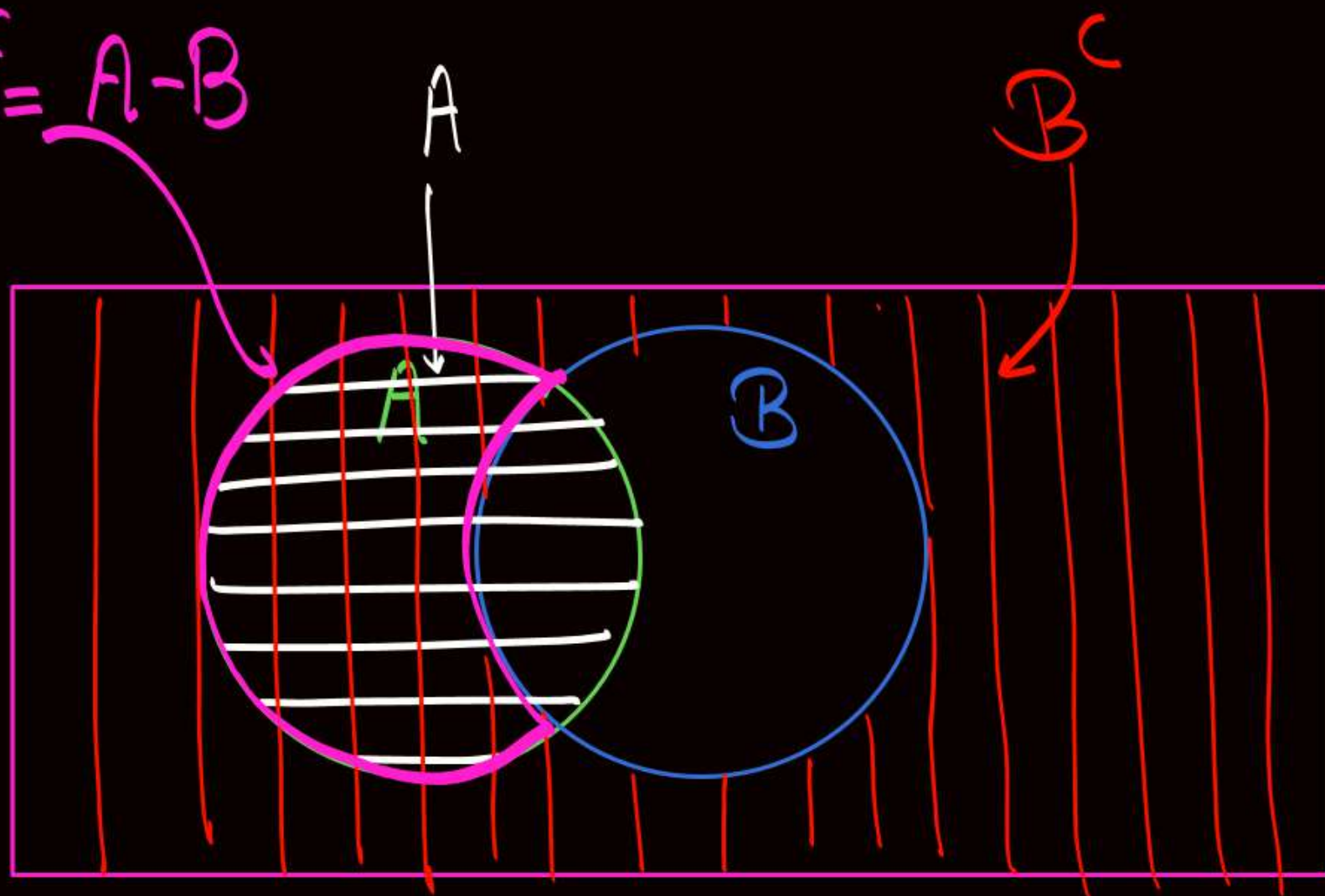
$$A \Delta B = B \Delta A$$



$$= (A \cup B) - (A \cap B)$$

$$A \Delta B = (A - B) \cup (B - A)$$

$$A \cap B^c = A - B$$





Topic : Properties of Set Operations

→ 1. Idempotent:

a. $A \cap A = A$

b. $A \cup A = A$

→ 2. Identity:

a. $A \cup \emptyset = A$

b. $A \cap U = A$



Topic : Properties of Set Operations

3. Domination:

a. $A \cap \emptyset = \emptyset$

b. $A \cup U = U$



Topic : Properties of Set Operations

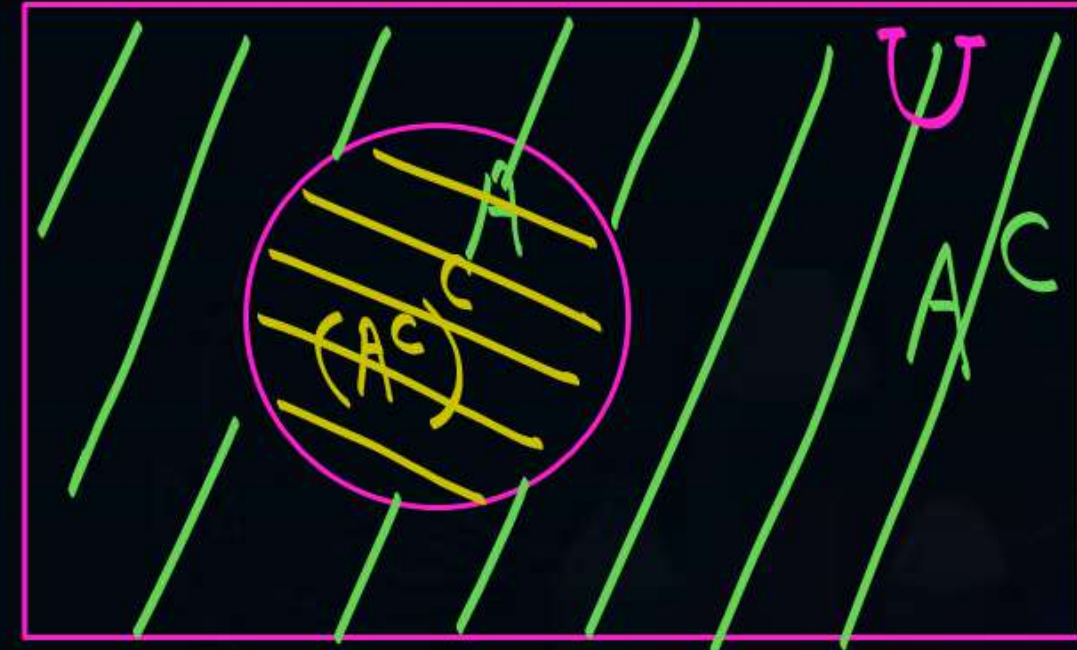
✓ 4. Complementation:

a. $A \cup A^c = U$

b. $A \cap A^c = \emptyset$

5. Double Complement:

a. $(A^c)^c = A$





Topic : Properties of Set Operations

6. Commutative

a. $A \cup B = B \cup A$

b. $A \cap B = B \cap A$

7. Associative

a. $A \cup (B \cup C) = (A \cup B) \cup C$

b. $A \cap (B \cap C) = (A \cap B) \cap C$



Topic : Properties of Set Operations

8. Absorption $A \cap B \subseteq A$ and $A \cap B \subseteq B$

a. $A \cup (A \cap B) = A$ ✓

b. $A \cap (A \cup B) = A$ ✓

$$A \cup B \supseteq A$$

9. DeMorgan's

a. $(A \cup B)^c = A^c \cap B^c$

b. $(A \cap B)^c = A^c \cup B^c$

$$(A^c \cup B^c)^c = A \cap B$$

$$(A^c \cap B^c)^c = A \cup B$$



Topic : Properties of Set Operations

$$\cup \equiv +$$

$$\cap \equiv \cdot$$

10. Distributive

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

b. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

$$A \cdot (B + C) = (A \cdot B) + (A \cdot C)$$

H.W Problem Solution



Q1. For a set A , the power set of A is denoted by 2^A . If $A = \{5, \{6\}, \{7\}\}$

which of the following options are true?

(MSQ)

✓ 1. $\emptyset \in 2^A$

✓ 2. $\emptyset \subseteq 2^A$

✓ 3. $\{5, \{6\}\} \in 2^A$

✗ 4. $\{5, \{6\}\} \subseteq 2^A$

Not the elements from 2^A

$P(A) = \{ \emptyset, \{5\}, \{\{6\}\}, \{\{7\}\}, \{5, \{6\}\}, \{5, \{7\}\}, \{\{6\}, \{7\}\}, \{5, \{6\}, \{7\}\} \}$

H.W. Problem

Solution



$$S = \{1, 2\}$$

$$P(S) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\} \quad |P(S)| = 2^2 = 4$$

$$P(P(S)) = \{\emptyset, \{\emptyset\}, \{\{1\}\}, \{\{2\}\}, \{\{1, 2\}\}, \{\emptyset, \{1\}\}, \{\emptyset, \{2\}\}, \{\emptyset, \{1, 2\}\}, \{\{1\}, \{2\}\}, \{\{1\}, \{1, 2\}\}, \{\{2\}, \{1, 2\}\}, \{\emptyset, \{1\}, \{2\}\}, \{\emptyset, \{1\}, \{1, 2\}\}, \{\emptyset, \{2\}, \{1, 2\}\}, \{\{1\}, \{2\}, \{1, 2\}\}\}$$

$|P(P(S))| = 16$

Q2. Let $P(S)$ denote the power set of a set S . Which of the following is always true?

~~A.~~ $P(P(S)) = P(S)$

B. $P(S) \cap P(P(S)) = \{\emptyset\}$

C. $P(S) \cap S = P(S)$

D. $S \notin P(S)$

(i) $\emptyset \in P(S)$ (True)

(ii) $\emptyset \in P(P(S))$ (True)

Therefore $\emptyset$ is guaranteed to be an element of $P(S) \cap P(P(S))$

$$P(S) \cap P(P(S)) = \{\emptyset\}$$

Statement 'B' may be true but not always

H.W. Problem Solution



$$S = \{ \underline{1}, \{1\} \}$$

$$P(S) = \{ \emptyset, \{1\}, \{\{1\}\}, \{1, \{1\}\} \}$$

Q2. Let $P(S)$ denote the power set of a set S . Which of the following is always true?

$$P(P(S)) = \{ \emptyset, \{ \emptyset \}, \{ \{1\} \}, \{ \{1, \{1\}\} \}, \{ \{1, \{1\}\} \} \}$$

~~A.~~ $P(P(S)) = P(S)$

~~B.~~ $P(S) \cap P(P(S)) = \{ \emptyset \}$

~~C.~~ $P(S) \cap S = P(S)$

~~D.~~ $S \notin P(S)$

$$P(S) \cap P(P(S)) = \{ \emptyset, \{ \{1\} \} \} \neq \{ \emptyset \}$$

$$|A| = 3 \quad \& \quad |B| = 5$$

$$\text{max size of } A \cap B = 3$$

Statement B may be false

$\hookrightarrow S \in P(S)$ is always true



Topic : Multi-set

An unordered collection of elements/objects in which an element may appear more than once is called a multi-set.

$$\text{eg. } A = \{a, a, b, c, b, b, d, d\}$$

$$= \{a, a, b, b, b, c, d, d\}$$

Multiplicity
of corresponding
elements

$$= \{2.a, 3.b, 1.c, 2.d\}$$

* Let,

Multiset $P = \{m_1 \cdot a_1, m_2 \cdot a_2, m_3 \cdot a_3, \dots, m_k \cdot a_k\}$
where m_i is the multiplicity of element a_i

and let,

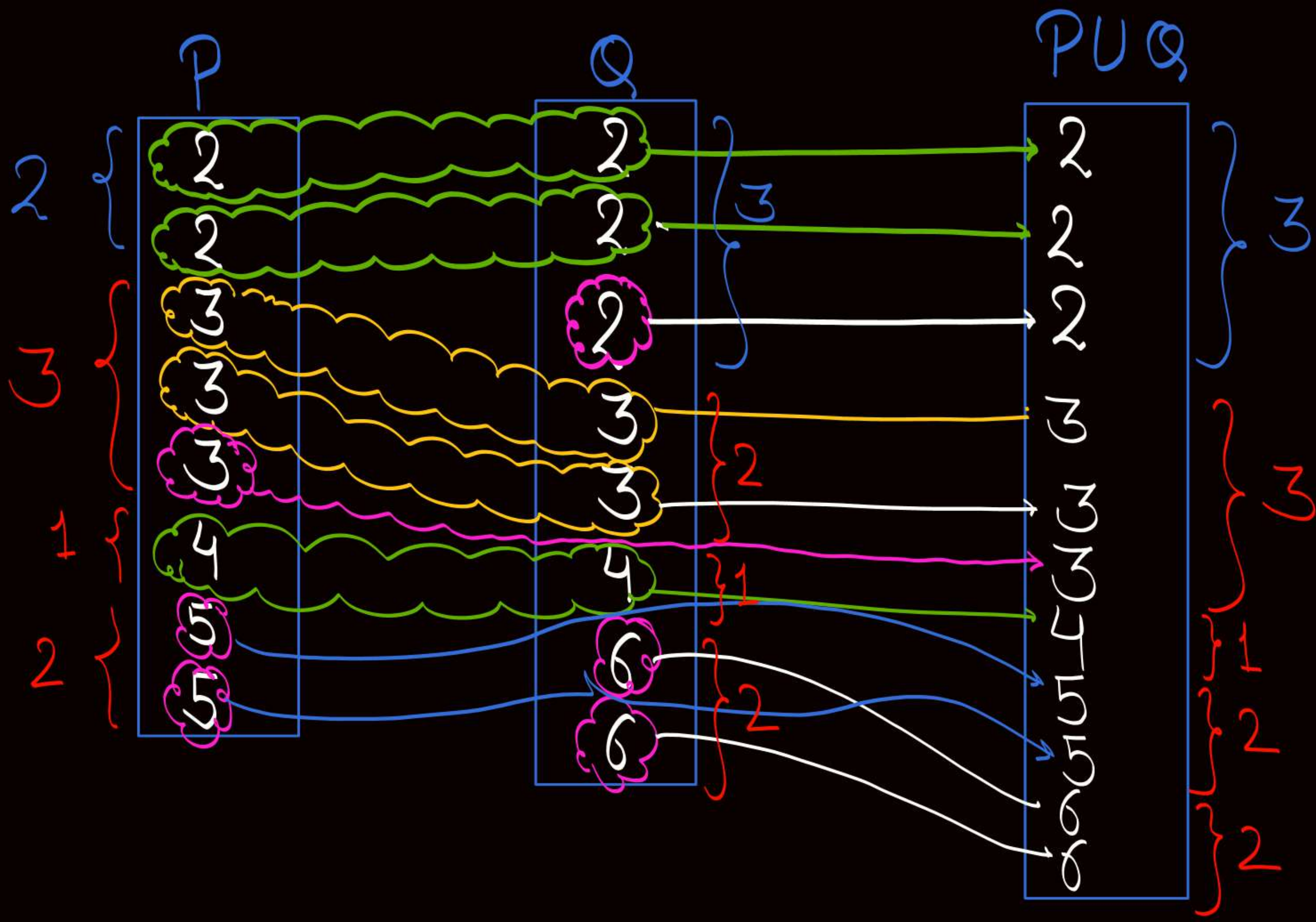
Multiset $Q = \{n_1 \cdot a_1, n_2 \cdot a_2, n_3 \cdot a_3, \dots, n_k \cdot a_k\}$
where n_i is the multiplicity of element a_i

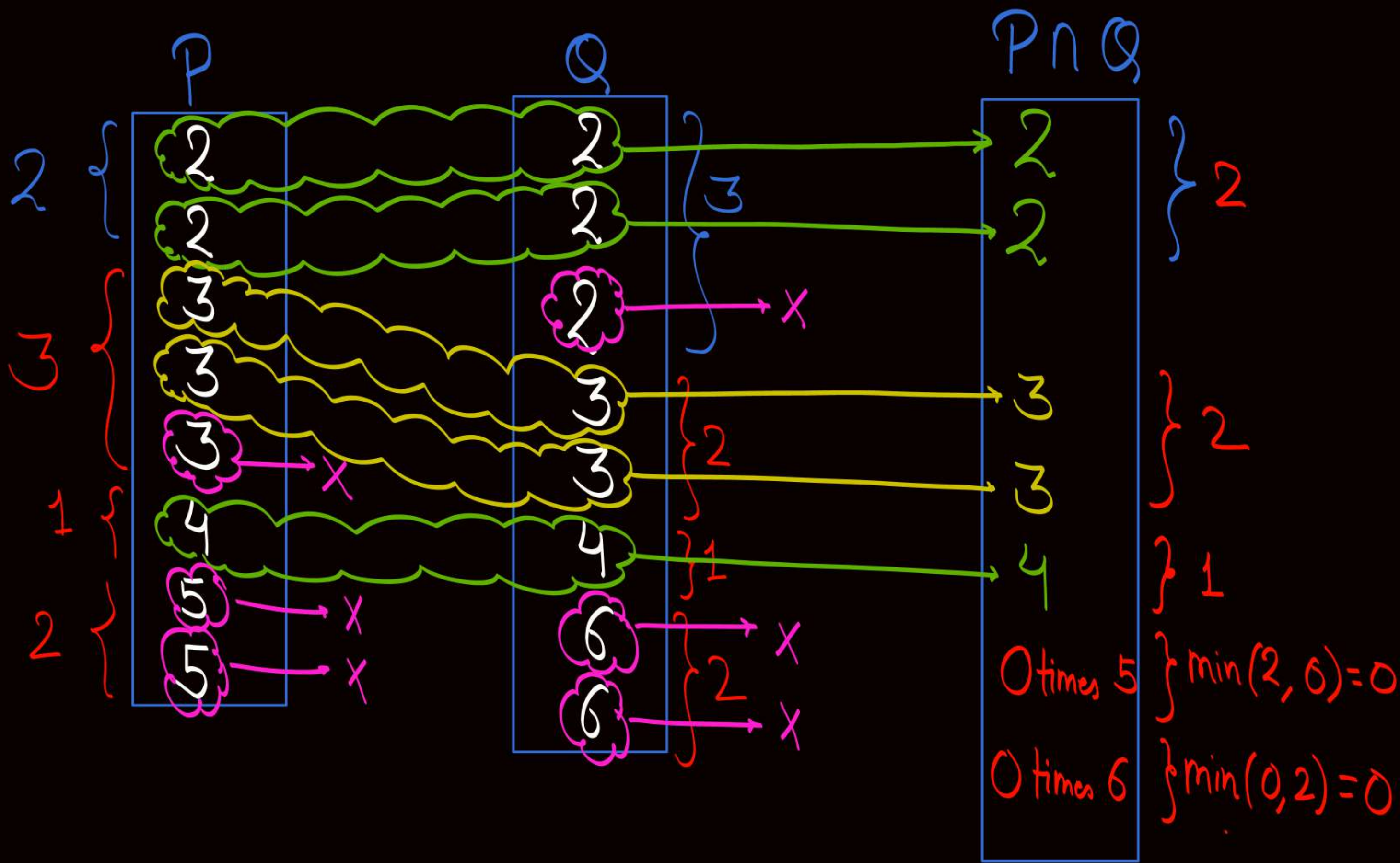
Q: Find multiplicity of a_i in,

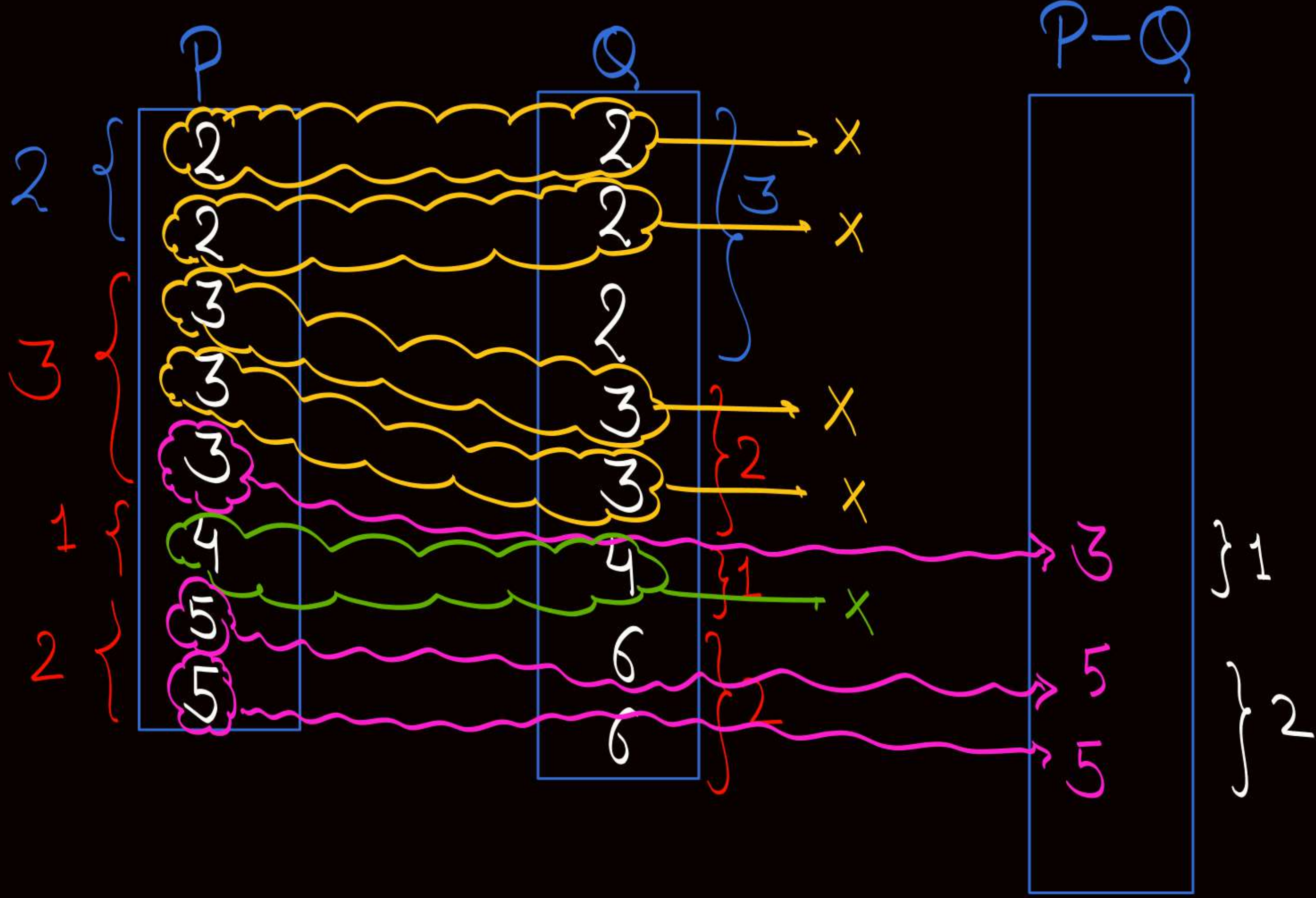
(i) $P \cup Q$

(ii) $P \cap Q$

(iii) $P - Q$







* Let,

Multiset $P = \{m_1 \cdot a_1, m_2 \cdot a_2, m_3 \cdot a_3, \dots, m_k \cdot a_k\}$
where m_i is the multiplicity of element a_i

and let,

Multiset $Q = \{n_1 \cdot a_1, n_2 \cdot a_2, n_3 \cdot a_3, \dots, n_k \cdot a_k\}$
where n_i is the multiplicity of element a_i

- (i) multiplicity of a_i in $P \cup Q = \max(m_i, n_i)$
- (ii) multiplicity of a_i in $P \cap Q = \min(m_i, n_i)$
- (iii) multiplicity of a_i in $P - Q = \begin{cases} (m_i - n_i), & \text{if } m_i > n_i \\ 0 & , \text{ otherwise} \end{cases}$

Q:- Let $A = \{1, 2, 3, 4, \dots, n\}$ {i.e. $|A| = n$ }

How many multi-sets are possible from the elements of set A. { Any element of set A can be used any number of times in the multi-set }

Soluⁿ.

$$M_1 = \{1\} \quad \text{size of } M_1 = 1$$

$$M_2 = \{1, 1\} \quad \text{size of } M_2 = 2$$

$$M_3 = \{1, 1, 1\} \quad \text{size of } M_3 = 3$$

$$\vdots$$
$$M_{100} = \{1, 1, 1, \dots, 1\} \quad \text{size of } M_{100} = 100$$

100 times '1'

$$\vdots$$
$$M_{\infty} = \{1, 1, 1, \dots\} \quad \text{size of } M_{\infty} = \text{infinite}$$

If we do not restrict the size of the multiset, then infinite multisets can be formed using a single element

Q:- Let $A = \{1, 2, 3, 4, \dots, n\}$ ✓

How many multisets of size = 4 can be formulated using the elements of set A , such that at least one element of set A appears exactly twice in each such multiset.

$A = \{1, 2, 3, \dots, n\}$ Based on the statement given in the question, two types of multi-set are possible

a, b, c ✓

a, a, b, c
 b, b, a, c
 c, c, a, b

$$\{a, a, b, c\} = \{a, b, c, a\}$$

$$\{a, a, b, b\} = \{a, b, b, a\}$$

$$\left(\binom{n}{c_1} * 1 * \binom{n-1}{c_1} * 1 * \binom{n-2}{c_1} * 1 \right) = \frac{n(n-1)(n-2)}{2}$$

$$\rightarrow \left(\frac{\binom{n}{c_1} * 1 * \binom{n-1}{c_1} * 1}{2} \right) = \frac{n(n-1)}{2}$$

$$\left(\binom{n}{c_1} * 1 * \binom{n-1}{c_2} * 1 \right) = \frac{n * (n-1)(n-2)}{2}$$

(or)

$$\binom{n}{c_2} * 1 = \frac{n(n-1)}{2}$$

write each one of them twice

$$\binom{n}{c_3} * 3 * \binom{n-1}{c_1} * 1 * 1 = \binom{n}{c_3} * 3 = \frac{n(n-1)(n-2)}{3 \cdot 2 \cdot 1} * 3$$

Total no. of required multisets = Multisets using Case ① or Multisets using Case ②

$$= \frac{n(n-1)(n-2)}{2} + \frac{n(n-1)}{2}$$

$$= \boxed{\frac{n(n-1)^2}{2}} \quad \underline{\underline{\text{Ans}}}$$

Q:- Let $A = \{1, 2, 3, 4, \dots, n\}$ ✓

H.W. How many multisets of size = 4 can be formulated using the elements of set A . { i.e., no other restriction apart from size of multiset = 4 }



2 mins Summary



✓ {
Topic

Set operations and properties of set operations

✓
Topic

Multi-set

Topic

Cartesian product

Topic

Relation

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THANK - YOU